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Bijon Guha

AI/ML Specialist – ZENSAR

Portfolio Website
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8+ years of expertise in designing and shipping AI/ML systems, leading teams across *Healthcare, Manufacturing,* and *Smart Automation.* Experience spans **Generative AI and agentic systems**, multi-tenant SaaS platforms, scalable AI pipelines, and low-latency deployment using LLM orchestration frameworks, the NVIDIA stack, Docker, and classical AI/ML.

Education

Integrated Master of Science, Applied Mathematics — **Jun 2013 – May 2018**
Indian Institute of Technology, Roorkee

Experience

Senior Technical Specialist — **Mar 2025 – Present**
Zensar Technologies — Bengaluru, India

- **Lead GenAI/ML Engineer — Compass Digital, USA *(Client)***
  - Architected the AI backend of ***Heidi***, a multilingual, multi-capability conversational AI assistant for Compass, cutting routine ticket load by **60%** across **500+ healthcare facilities** in the USA.
  - Built the *Talk-To-My-Data* path on **Snowflake Cortex Analyst** handling **1M+ transactions/day**. Enforced role-based access control via **Row-Level Security** (*first of its kind in the org*), bringing data leaks to **zero** across table collections spanning multiple business units and roles.
  - Pitched and led creation of a formal **evaluation framework** for Heidi's evolving multi-capability system; expanded into **Agentic QA**, where an agent QAs Heidi alongside humans to regression-test existing abilities and quantify the impact of each new ability on defined metrics.
  - Extended Compass's meal recommender from a traditional ML item recommender to an **LLM-based** coherent, palatable meal suggester, raising the patient satisfaction index by **13%**. Performed exhaustive prompt and state-graph optimization to meet strict latency requirements.
  - Secured **1st position** in the Compass Hackathon for building and demoing ***TrainBrain***; the idea was approved as an official Compass product. Subsequently awarded **Top Talent, Zensar (2026–2028)**.
  - *Skillset: AWS Bedrock, Azure OpenAI, LangChain, LangGraph, Snowflake Cortex, RAG, Text-to-SQL, Agentic AI*

Lead Machine Learning Engineer — **Mar 2022 – Mar 2025**
Siemens Technology and Services Private Limited — Bengaluru, India

- **Lead Developer — GenAI Marketplace**
  - Conceptualized and led development of a **GenAI marketplace** for internal Siemens applications, significantly enhancing organization-wide productivity.
  - Developed ***Jira-Copilot*** and ***EstiMATE***, deployed on AWS at MVP scale with **450+ active users** (until Q1 2025). Integrated with the Atlassian Jira dashboard to automate story updates.
  - *Skillset: OpenAI, Azure, LangChain, ChromaDB, Prompt Engineering, Advanced RAG, AI Agents, LangGraph*
- **Lead Developer — CKAN**
  - Developed extensions for an open-source data platform supporting European universities.
  - Integrated AWS for data uploads and **Elasticsearch** for efficient search.
  - *Skillset: AWS, Elasticsearch, Python, Concurrent Processing, CI, CKAN*
- **Tech Lead — Automotive Anomaly Detection AI Platform**
  - Authored HLD/LLD documents and designed the end-to-end system for a multi-tenant SaaS platform.
  - Developed MQTT, Postgres, DL inference, and system monitoring services.
  - Processed, stored, and purged **32 images in under 5 seconds** (handling **10 TB+ data/day**) at high precision.
  - *Skillset: Low Latency Deployment, NVIDIA Triton, DVC, Concurrent Processing, Vision Transformers*
- **Drone-Vision AI Service Deployment Lead**
  - Established a scalable deep learning service template covering **10+ use cases** on AWS cloud.
  - Developed AWS Lambda services for maintenance tasks and contributed to CloudFormation templates.
  - *Mentored junior developers* transitioning from Jupyter notebooks to production-grade code.
  - *Skillset: MM-Detection, Tiny Object Detection, MLflow, SAHI, RabbitMQ, Transformers, Kubectl*

Machine Learning Engineer — **Mar 2021 – Feb 2022**
Cargill Business Services — Bengaluru, India

- **Garry — Label OCR System**
  - Developed a deep learning microservice for label OCR and entity extraction from human-captured mobile and iPad images with real-world perturbations.
  - Created performance benchmarks across **PaddleOCR, EasyOCR,** and **KerasOCR**, combined with downstream NLP extraction.

– Fine-tuned PaddleOCR models to reduce **CER and WER**, and applied **quantization (QAT, PTQ)** for edge-device deployment.
– Deployed a LabelImg microservice for image annotation and added **MLflow** for model tracking.
– *Skillset: Python, PaddleOCR, FastAPI, MLflow, spaCy, Label Studio, CI/CD, Hugging Face, AWS SageMaker Serving*

Senior Data Scientist **Jul 2020 – Mar 2021**
QpiAI (SIEMENS) Bengaluru, India

- **Visual Quality Inspection System**
  – Annotated data and trained tiny-object detection models (**Faster R-CNN, SSD**) with ResNet and VGGNet backbones to detect electrical components from electrical wiring plans.
  – Countered scarce industrial data with a synthetic augmentation workflow built on Siemens Capital Design, and modified loss functions (*classes were not mutually exclusive*), achieving a **mAP of 80+**.
  – Trained and fine-tuned **CRAFT OCR** for detecting and extracting disc codes at a drum-disk manufacturing facility. Annotated data, versioned models with MLflow, and implemented homography for entity extraction. Deployed *air-gapped* on an edge device for factory-floor use via Docker orchestration.
  – *Skillset: Python, CRAFT OCR, Image Processing, Synthetic Data Generation*
- **Model Deployment Microservice**
  – Built ***QpiAI-Pro Deploy*** to manage the deployment and scaling of AI models.
  – Implemented **quantization, pruning,** and **distillation**; configured OpenVINO and NNCF.
  – *Skillset: Quantization, Pruning, OpenVINO, NNCF, Docker, Code Binarization*

Data Scientist **Jul 2018 – Jul 2020**
Bosch Bengaluru, India

- **Computer Vision Team**
  – *Image Classification* — Built production image classifiers on aggregated multi-source datasets; fine-tuned ResNet backbones in PyTorch with one-cycle LR scheduling and weight decay, reaching **98% top-1 accuracy** in 200 epochs vs. 80% for the baseline schedule.
  – *Object and Face Detection* — Delivered single-shot detection pipelines using **YOLO** (Darknet backbone) and **RetinaFace** (ResNet/MobileNet backbones); retrained YOLO on a custom-annotated domain dataset, improving mAP from **44% to 82%** while sustaining **10 FPS** on Intel Xeon CPU.
  – *Object Counting (multi-approach)* — Owned an end-to-end counting system across three methods: classical CV (Hough circles, blob detection, morphological separation), detection-based counting benchmarked across SSD-MobileNet, SSD-ResNet and Faster R-CNN-NASNet via the TensorFlow Object Detection API, and density-map regression with a U-Net trained from scratch on biological cell imagery. Built an in-house point-annotation tool for ground truth and authored the comparative accuracy/latency report that drove final model selection.
  – *Biomedical Signal Classification* — Validated and hardened a production signal-classification model; parallelized Pandas preprocessing to cut training time **10x**, then packaged the model as a Python library deployed to a constrained edge device.
  – *Skillset: Transfer Learning, Image Processing, Real-time Inference, Semantic Segmentation, ResNet, Data Parallelization*
- **Handwritten Optical Character Recognition**
  – Designed an end-to-end OCR system for handwritten mathematical equations using deep learning.
  – Developed a plug-and-play Python package with Keras and TensorFlow for an *Intel India open-source project.*
  – *Skillset: OCR, DCNN, Weight Decay, Model Ensemble*

Publications

1. Indrajit Kar Sudipta Mukhopadhyay, B. G. A Dual Fine Grained Rotated Neural Network for Aerial Solar Panel Health Monitoring and Classification. *ICDMAI* (2023).
2. Nimai Chand Das Adhikari Bijon Guha, U. D. PiXelNet: A DL-Based method for Diagnosing Lung Cancer using histopathological images. *ACMLC* (2023).
3. Nimai Chand Das Adhikari Bijon Guha, A. A. Hawkeye: A concept for smart home surveillance system. *PSE* (2023).

Certifications

- **Claude Certified Associate — Foundations** *(Anthropic)* Aug 2026 – Aug 2027
  – Proctored certification for *client-facing and delivery practitioners*: applying Claude to business workflows, evaluating and refining model output, selecting the right products and models, and practising **responsible AI**.
  – *Skills validated: Prompt Engineering, Model Selection, Output Evaluation, Responsible AI, Workflow Integration, Knowledge Management, Claude Projects, Troubleshooting*
  – *Verify on Credly*

Awards & Achievements

2013 Secured **All India Rank 4405** in JEE-Advanced (>99th percentile)

2015 Passed *"CT3 — Probability and Statistics"* by the Institute of Actuaries of India

2018 Recipient of the **INSPIRE Scholarship** for Higher Education, Department of Science and Technology, India

2019 Featured as **Fresher Best Performer**, EDS Team, Bosch

2021 Received the **Innovation Award**, Intelligent Automation Team, Cargill

2023 **Promoted to Lead ML Engineer** within a year at Siemens

2024 **Top Winner**, XED'24 Innovation Challenge, Siemens

2026 **Winner**, AI Summit Hackathon, Compass Digital USA

2026 **Emerging Top Client Focus Talent**, Zensar

2026 Awarded **Top Talent, Zensar** *(valid for 2 years)*